

CSCI E-278 - Assignment 3

Problem Set Solutions

Question 1

An analyst estimates the volatility of a stock using its historical returns. An options trader instead starts with the current market price of an option and determines the value of volatility that would make the option-pricing model reproduce that market price. What type of volatility is each person using? Explain why the second measure is called implied volatility and how its source of information differs from historical volatility.

Answer. The analyst is using **historical (realized) volatility**, which is estimated from past stock returns. The options trader is using **implied volatility**. It is called implied because volatility is not observed directly; instead, the trader takes the option's market price and solves backward for the volatility input that makes an option-pricing model match that price. Historical volatility therefore summarizes past realized price movements, while implied volatility is extracted from current option prices and reflects the market's forward-looking expectations and risk pricing.

Question 2

Suppose ordinary lag-1 differencing of a stock-price series produces a return series with an ADF statistic around -21 and a p-value near zero. A fractally differenced version of the same series instead has an ADF statistic around -3.5 while still rejecting the null hypothesis of non-stationarity. Both series appear stationary. Based on the lecture, why might the first series be less useful for prediction even though it is more strongly stationary? What is fractal differencing trying to preserve that ordinary differencing may remove?

Answer. Ordinary differencing may make the series extremely stationary by removing not only the non-stationary trend but also useful long-memory structure and predictive information contained in the original price series. The much more negative ADF statistic does not automatically make the transformed series better for forecasting. **Fractal (fractional) differencing** tries to use the minimum amount of differencing needed to achieve stationarity while preserving as much memory, dependence, and economically useful information from the original series as possible.

Question 3

Two stocks each have non-stationary price series, but an analyst finds coefficients alpha and beta such that the spread $S_t = Y_t - \beta X_t - \alpha$ is stationary and tends to revert toward a stable level. What does this tell the analyst about the relationship between the two stocks, and how is this concept different from simply observing that their returns are correlated? Based only on the high-level trading intuition introduced in this lecture, how might a temporary divergence between the two stocks create an opportunity?

Answer. The two price series are **cointegrated**: although each price is individually non-stationary, a particular linear combination of them is stationary, indicating a stable long-run equilibrium relationship. This is stronger and different from simple return correlation, which only measures how two variables tend to move together and does not imply a mean-reverting relationship between their price levels. If the spread temporarily moves far from its typical level, a pairs-trading intuition is to position for convergence - buying the relatively underpriced side and/or selling the relatively overpriced side - expecting the spread to revert.

Question 4

A researcher has access to hundreds of possible stock features and proposes using the same complete feature set for every company, regardless of whether the company is in technology, financials, energy, or another sector. Explain why the lecture's 'one size does not fit all' principle makes this approach problematic. What does it mean for features to be relatively orthogonal, and why can removing highly correlated features improve the resulting model?

Answer. Different industries are driven by different economics, so a feature that is informative for a bank may be irrelevant or misleading for a technology or energy company. Using every feature everywhere can add noise, redundancy, instability, and overfitting. Relatively **orthogonal** features contain distinct information rather than largely duplicating one another. Removing highly correlated features reduces multicollinearity and the effective dimensionality of the problem, making parameter estimates more stable and allowing the model to focus on independent sources of predictive information.

Question 5

A machine-learning model fits its training data almost perfectly, but its predictions are much worse when evaluated on a separate test dataset. What problem does this pattern suggest? Explain how regularization can help address it, including what happens to the weights of less useful features and why reducing the effective feature space can improve performance on new data.

Answer. This pattern indicates **overfitting**: the model has learned training-specific noise or overly complex relationships that do not generalize. Regularization penalizes model complexity and discourages unnecessarily large coefficients. Less useful features have their weights shrunk toward zero; with methods such as L1 regularization, some coefficients can become exactly zero. This effectively reduces the feature space and model variance, making the model less sensitive to noise and often improving performance on unseen data.